# Lecture 7

- Kalman Filter derivation

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Recall from last time:

## Kalman Filter Algo.

Given:

model parameters $A_t, B_t, C_t, D_t, Q_t, R_t$
and prior $X_0 \sim N(\mu_{0|0}, \Sigma_{0|0})$

**predict:**

mean: $\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$

cov: $\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$

$p(x_t|y_{1:t-1}) = N(\mu_{t|t-1}, \Sigma_{t|t-1})$

**update:**

mean: $\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1})$

cov: $\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$

where $K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$ ← Kalman Gain

Gives Bayesian posterior $p(x_t|y_{1:t}) = N(\mu_{t|t}, \Sigma_{t|t})$ for: (")

## Linear Gaussian SS Model

$$X_0 \sim N(\mu_{0|0}, \Sigma_{0|0})$$

$$X_{t+1} = A_t X_t + B_t u_t + W_t$$

$$W_t \sim GWN(0, Q_t)$$

$$Y_t = C_t X_t + D_t u_t + V_t$$

$$V_t \sim GWN(0, R_t)$$

(and more conditions to follow)

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recall the notation:

$\mu_{t|t} := E[X_t | Y_{1:t}]$

$\mu_{t|t-1} := E[X_t | Y_{1:t-1}]$

in general:

$\mu_{t|T} := E[X_t | Y_{1:T}]$

and $\Sigma_{t|t} := Cov(X_t | Y_{1:t})$

$= E[(X_t - \mu_{t|t})(X_t - \mu_{t|t})^T | Y_{1:t}]$

$\Sigma_{t|t-1} := Cov(X_t | Y_{1:t-1})$

$\Sigma_{t|T} := Cov(X_t | Y_{1:T})$

graphically)




let's prove the KF eqns

We start with update eqns.

Insight, these are actually Gaussian Ed. eqns in disguise.

Recall Gauss. Ed. eqns.

mean: $\mu_{x|y} = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y - \mu_y)$

cov: $\Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$

let's map over to KF notation

$\mu_{t|t} = E[X_t | Y_{1:t}] = E[X_t | \underbrace{Y_t}_{\text{new measurement}}, Y_{1:t-1}]$ ("like" $\mu_{x|y}$)

$\mu_{t|t-1} = E[X_t | Y_{1:t-1}]$ — prior ("like" $\mu_x$)

same for Cov:

$\Sigma_{t|t}$ "like" $\Sigma_{x|y}$, $\Sigma_{t|t-1}$ "like" $\Sigma_x$

define:

$\mu_{y_t} = E[Y_t | Y_{1:t-1}]$

$\Sigma_{y_t} = Cov(Y_t | Y_{1:t-1}) = E[(Y_t - \mu_{y_t}) \cdot (Y_t - \mu_{y_t})^T | Y_{1:t-1}]$

$\Sigma_{x_t y_t} = Cov(X_t, Y_t | Y_{1:t-1})$

$= E[(X_t - \mu_{t|t-1})(Y_t - \mu_{y_t})^T | Y_{1:t-1}]$

rewriting Gauss Est. eqns:

$$\mu_{t|t} = \mu_{t|t-1} + \Sigma_{x_t y_t}\Sigma_{y_t}^{-1}(y_t - \mu_{y_t})$$
$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{x_t y_t}\Sigma_{y_t}^{-1}\Sigma_{y_t x_t}$$

claim:

(i) $\mu_{y_t} = E[Y_t | Y_{1:t-1}] = C_t \mu_{t|t-1}$

(ii) $\Sigma_{x_t y_t} = Cov(X_t, Y_t | Y_{1:t-1}) = \Sigma_{t|t-1} C_t^T$

(iii) $\Sigma_{y_t} = Cov(Y_t | Y_{1:t-1}) = (C_t \Sigma_{t|t-1} C_t^T + R_t)$

Let's prove these claims.

(i) from L-G SS Model we have

$$Y_t = C_t X_t + D_t u_t + V_t$$
$$V_t \sim GWN(0, R_t)$$

$$\mu_{y_t} = E[Y_t | Y_{1:t-1}]$$
$$= E[C_t X_t + D_t u_t + V_t | Y_{1:t-1}]$$
$$= C_t \underbrace{E[X_t | Y_{1:t-1}]}_{\mu_{t|t-1}} + D_t \underbrace{E[u_t | Y_{1:t-1}]}_{\text{assume } u_t \text{ deterministic} = u_t} + \underbrace{E[V_t | Y_{1:t-1}]}_{0}$$

A1: assuming $V_t$ indep. $Y_{1:t-1}$

we get

$$\mu_{y_t} = C_t \mu_{t|t-1} + \cancel{D_t u_t}^{\,0}$$

A0: assuming $D_t = 0$ (if A0 not true, easy to modify KF accordingly)

$$\boxed{\mu_{y_t} = C_t \mu_{t|t-1}}$$ □

(ii)

$$\Sigma_{x_t y_t} = E[(X_t - \mu_{t|t-1})(Y_t - \mu_{y_t})^T | Y_{1:t-1}]$$

sub measurement eqn

$$(Y_t - \mu_{y_t}) = C_t X_t + V_t - C_t \mu_{t|t-1}$$

$$= E[(X_t - \mu_{t|t-1})(C_t(X_t - \mu_{t|t-1}) + V_t)^T | Y_{1:t-1}]$$

$$= \underbrace{E[(X_t - \mu_{t|t-1})(X_t - \mu_{t|t-1})^T | Y_{1:t-1}]}_{\Sigma_{t|t-1}} C_t^T + \underbrace{E[(X_t - \mu_{t|t-1}) V_t^T | Y_{1:t-1}]}_{0 \text{ (A2)}}$$

A2: assuming $X_t$ indep. $V_t$ given $Y_{1:t-1}$

we find

$$\boxed{\Sigma_{x_t y_t} = \Sigma_{t|t-1} C_t^T}$$ □

(iii)

$$\Sigma_{y_t} = E[(Y_t - \mu_{y_t})(Y_t - \mu_{y_t})^T | Y_{1:t-1}]$$

we saw

$$(Y_t - \mu_{y_t}) = C_t(X_t - \mu_{t|t-1}) + V_t$$

sub into above

$$= E[(C_t(X_t - \mu_{t|t-1}) + V_t)(C_t(X_t - \mu_{t|t-1}) + V_t)^T | Y_{1:t-1}]$$

$$= \underbrace{C_t E[(X_t - \mu_{t|t-1})(X_t - \mu_{t|t-1})^T | Y_{1:t-1}] C_t^T}_{\Sigma_{t|t-1}}$$
$$+ \underbrace{C_t E[(X_t - \mu_{t|t-1}) V_t^T | Y_{1:t-1}]}_{0 \text{ (A2)}}$$
$$+ \underbrace{E[V_t (X_t - \mu_{t|t-1})^T | Y_{1:t-1}] C_t^T}_{0 \text{ (A2)}}$$
$$+ \underbrace{E[V_t V_t^T | Y_{1:t-1}]}_{E[V_t V_t^T] \text{ (by A1)}}$$

$= R_t$ by def.

we have $\boxed{\Sigma_{y_t} = (C_t \Sigma_{t|t-1} C_t^T + R_t)}$ □

put it all together:

$$\mu_{t|t} = \mu_{t|t-1} + \underbrace{(\Sigma_{t|t-1} C_t^T)}_{\Sigma_{x_t y_t}} \underbrace{(C_t \Sigma_{t|t-1} C_t^T + R_t)^{-1}}_{\Sigma_{y_t}^{-1}} \times (y_t - \underbrace{(C_t \mu_{t|t-1})}_{\mu_{y_t}})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \underbrace{(\Sigma_{t|t-1} C_t^T)}_{\Sigma_{x_t y_t}} \underbrace{(C_t \Sigma_{t|t-1} C_t^T + R_t^{-1})}_{\Sigma_{y_t}} \times \underbrace{(C_t \Sigma_{t|t-1})}_{\Sigma_{y_t x_t}}$$

this is update step for KF! □

notice: Kalman gain:

$$K_t := \Sigma_{x_t y_t}\Sigma_{y_t}^{-1} = \Sigma_{t|t-1} C_t^T (C_t \Sigma_{t|t-1} C_t^T + R_t)^{-1}$$

Now the prediction step:

$$\mu_{t|t-1} = E[X_t | Y_{1:t-1}]$$

from SS model

$$X_t = A_{t-1} X_{t-1} + B_{t-1} u_{t-1} + W_{t-1}$$

$$W_{t-1} \sim GWN(0, Q_{t-1})$$

$$\mu_{t|t-1} = E[A_{t-1} X_{t-1} + B_{t-1} u_{t-1} + W_{t-1} | Y_{1:t-1}]$$

$$= A_{t-1} \underbrace{E[X_{t-1} | Y_{1:t-1}]}_{\mu_{t-1|t-1}} + B_{t-1} \overbrace{E[u_{t-1} | Y_{1:t-1}]}^{u_{t-1}} + \underbrace{E[W_{t-1} | Y_{1:t-1}]}_{=0}$$

$$\boxed{\mu_{t|t-1} = A_{t-1} \mu_{t-1|t-1} + B_{t-1} u_{t-1}}$$ □

A3: assume $W_t$ indep of $Y_{1:t}$

$$\Sigma_{t|t-1} = E[(X_t - \mu_{t|t-1})(X_t - \mu_{t|t-1})^T | Y_{1:t-1}]$$

plug in SS model

⋮ algebra

A4: $X_t$ indep of $W_t$ given $Y_{1:t-1}$

$$= A_{t-1} \Sigma_{t-1|t-1} A_{t-1}^T + \text{cross terms} + \underbrace{E[W_{t-1} W_{t-1}^T | Y_{1:t-1}]}_{= Q_{t-1} \text{ by A3}}$$

we get

$$\boxed{\Sigma_{t|t-1} = A_{t-1} \Sigma_{t-1|t-1} A_{t-1}^T + Q_{t-1}}$$ □

which is the KF prediction step.